

Md Khayrul Islam

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EDUCATION

Lehigh University

Ph.D. in Mechanical Engineering; GPA: 3.84/4.00

- Awarded P.C. Rossin College of Engineering **fellowship**.

Pennsylvania, USA

Jan 2021 – May 2025

Lehigh University

M.S. in Mechanical Engineering (awarded en route to Ph.D.); GPA: 3.84/4.00

Pennsylvania, USA

Jan 2021 – May 2023

Thesis Title: Predicting Drug Loading in Extracellular Vesicles through Coarse-Grained Molecular Dynamics Simulation.
Published in **Phys. Chem. Chem. Phys.**

IBM

Professional Certificate in Data Science

Online

Aug 2024

- Built competencies in Python, machine learning, SQL, and data visualization, culminating in a capstone project.

SKILLS

Programming: Python , MATLAB , Bash , SQL 

ML & Computer Vision: PyTorch , ViT , ONNX , OpenCV , Distributed Training , GPU Profiling 

Simulation: COMSOL , EspressoMD , ANSYS , LAMMPS , GROMACS , VMD , SolidWorks 

Miscellaneous: Docker , Git , High-Performance Computing , SLURM , MLflow , Hydra , Cloud Computing 

WORK AND RESEARCH EXPERIENCE

Lawrence Livermore National Laboratory

Postdoctoral Researcher – Machine Learning and Computer Vision

Livermore, CA

Sep 2025 – Present

- Sole developer of SNAP (Smart Neural Annotation Pipeline), an ML-guided image segmentation and annotation tool serving as the data backbone for NIF's (National Ignition Facility) transition to ML-based CV pipelines, supporting NNSA mission-critical capabilities at LLNL.
- Developing advanced ML and computer vision algorithms for identifying and tracking defects in NIF laser optics at the Computational Engineering Division.
- Conducting research on optical metrology analysis, automated decision-making frameworks, and predictive modeling to enhance optics performance and longevity, in collaboration with multidisciplinary teams of scientists and engineers.

University of Washington

Research Affiliate, Department of Electrical & Computer Engineering

Seattle, WA

Jul 2025 – Aug 2025

- Formalized a research collaboration on ML architecture development and hardware-aware optimization for biomedical image analysis.
- Contributed to FPGA-based acceleration strategies and system-level design for ultra-low-latency inference.

Lehigh University

Graduate Research Assistant

Bethlehem, PA

Jan 2021 - May 2025

- Collaborated in a multidisciplinary team to publish journal papers in computational biology and machine learning, featured in prestigious journals including Nature Communications, ACS, and RSC series, with 250+ citations.
- Supervised 4 undergraduate and 1 master's students; led 5 multidisciplinary group projects involving students from Bioengineering, Computer Science, and Mechanical Engineering.
- Administered a \$500K NSF-funded **high-performance computing (HPC) cluster**. Engineered initial setup and optimized on-demand resource allocation, achieving production-readiness for multidisciplinary research applications.

PROJECTS

Variable-Granularity Tokenization for Detection

Jan 2026 – Present

Designed adaptive patch tokenization for Vision Transformer object detection on tiny objects, allocating fine attention tokens to structurally active image regions and coarsening background to cut compute 22.8% and peak memory 57%, while setting state-of-the-art 48.0 AP on VisDrone and 37.26 AP on AI-TOD using PyTorch and MMDetection.

ML-Guided Segmentation & Annotation Pipeline

Sep 2025 – Present

Built a training-free, memory-augmented segmentation and annotation pipeline that turns the Segment Anything Model into a domain-specialized annotator for scarce, heterogeneous scientific imaging, reaching 91% mask acceptance from just 3 seed images while cutting annotator effort roughly 80%, using PyTorch and zero-shot segmentation with no fine-tuning at all.

Sub-Millisecond Cell Sorting

May 2024 – Dec 2025

Designed a label-free cell-classification framework for sub-millisecond microfluidic sorting, training a ResNet50 teacher reaching 98% accuracy and using knowledge distillation to compress it into a 5,682-parameter student, a 5000-fold reduction, while automating single-cell detection with YOLOv5 at 98% accuracy, removing 300+ hours of manual cropping, using PyTorch throughout. **Digital Discovery**

Multiplex Image ML

May 2023 – Dec 2024

First-authored MIML, a multimodal framework fusing label-free bright-field cell images with per-cell biomechanical measurements via a CNN feature-fusion architecture built in PyTorch, reaching 98.3% accuracy and 96.3% F1 on tumor-cell classification, outperforming image-only baselines, with Grad-CAM interpretability and successfully demonstrated transfer learning to other cell types. [Microsystems & Nanoengineering](#)

UsTaxes Open-Source Contribution

Jan 2026 – Mar 2026

Refactored PDF processing pipeline to schema-driven named-field mapping, eliminating fragile positional indexing. Built automated schema extraction and JSON validation framework; integrated CI checks to enforce consistency and prevent breakage from external PDF changes; applied system to support Y2025 tax forms. [GitHub](#)

PUBLICATIONS

1. **Khayrul Islam**, Mehedi Hasan, and Yaling Liu. Physics-guided surrogate modeling for machine learning-driven DLD design optimization. *Digital Discovery*, 2026. [In press]
2. Mehedi Hasan, Ishtiaq Murshed, **Khayrul Islam**, and A. K. M. Masud. Toward unified interphase engineering: The solid-electrolyte interphase in batteries and supercapacitors. *Journal of Materials Chemistry A*, 14(31):20009–20057, 05 2026
3. **Khayrul Islam**, Ryan F. Forelli, Jianzhong Han, Deven Bhadane, Jian Huang, Joshua C. Agar, Nhan Tran, Seda Oğrenci, and Yaling Liu. Real-time cell sorting with scalable in situ FPGA-accelerated deep learning. *Digital Discovery*, 5:254–265, 2026
4. **Khayrul Islam**, Ratul Paul, Shen Wang, Yuwen Zhao, Partho Adhikary, Qiying Li, Xiaochen Qin, and Yaling Liu. MIML: Multiplex image machine learning for high precision cell classification via mechanical traits within microfluidic systems. *Microsystems & Nanoengineering*, 11(1):43, 2025
5. Yue Wu, **Khayrul Islam**, Yaling Liu, and Anand Jagota. Microdroplet resuspension off surfaces. *Langmuir*, 40(52):27206–27215, 2024
6. Mehedi Hasan, **Khayrul Islam**, and AKM Masud. Tailoring polyamide nanocomposites: The synergistic effects of SWCNT chirality and maleic anhydride grafting. *ACS Applied Engineering Materials*, 2024
7. Yuwen Zhao, Yue Wu, **Khayrul Islam**, Ratul Paul, Yuyuan Zhou, Xiaochen Qin, Qiying Li, and Yaling Liu. Microphysiologically engineered vessel-tumor model to investigate vascular transport dynamics of immune cells. *ACS Applied Materials & Interfaces*, 16(18):22839–22849, 2024
8. Shen Wang, Jianzhong Han, Jingru Huang, **Khayrul Islam**, Yuheng Shi, Yuyuan Zhou, Dongwook Kim, Jane Zhou, Zhaorui Lian, Yaling Liu, et al. Deep learning-based predictive classification of functional subpopulations of hematopoietic stem cells and multipotent progenitors. *Stem Cell Research & Therapy*, 15(1):74, 2024
9. Yue Wu, Yuwen Zhao, **Khayrul Islam**, Yuyuan Zhou, Saeed Omid, Yevgeny Berdichevsky, and Yaling Liu. Acoustofluidic engineering of functional vessel-on-a-chip. *ACS Biomaterials Science & Engineering*, 9(11):6273–6281, 2023
10. Ratul Paul, Yuwen Zhao, Declan Coster, Xiaochen Qin, **Khayrul Islam**, Yue Wu, and Yaling Liu. Rapid prototyping of high-resolution large format microfluidic device through maskless image guided in-situ photopolymerization. *Nature Communications*, 14(1):4520, 2023
11. Anshu Raj, Sk Md Ahnaf Akif Alvi, **Khayrul Islam**, Mohammad Motalab, and Shuozhi Xu. An atomistic study of the tensile deformation of carbon nanotube–polymethylmethacrylate composites. *Polymers*, 15(13):2956, 2023
12. Yue Wu, Yuwen Zhao, Yuyuan Zhou, **Khayrul Islam**, and Yaling Liu. Microfluidic droplet-assisted fabrication of vessel-supported tumors for preclinical drug discovery. *ACS Applied Materials & Interfaces*, 15(12):15152–15161, 2023
13. **Khayrul Islam**, Meghdad Razizadeh, and Yaling Liu. Coarse-grained molecular simulation of extracellular vesicle squeezing for drug loading. *Physical Chemistry Chemical Physics*, 25(17):12308–12321, 2023
14. Yue Wu, Yuyuan Zhou, Ratul Paul, Xiaochen Qin, **Khayrul Islam**, and Yaling Liu. Adaptable microfluidic vessel-on-a-chip platform for investigating tumor metastatic transport in bloodstream. *Analytical Chemistry*, 94(35):12159–12166, 2022
15. Mehdi Nikfar, Ratul Paul, **Khayrul Islam**, Meghdad Razizadeh, Anand Jagota, and Yaling Liu. Respiratory droplet resuspension near surfaces: Modeling and analysis. *Journal of Applied Physics*, 130(2), 2021
16. **Khayrul Islam**, Sourav Saha, and AKM Masud. Molecular dynamics simulation of the mechanical properties of CNT-polyoxymethylene composite with a reactive forcefield. *Molecular Simulation*, 46(5):380–387, 2020

CONFERENCE PRESENTATIONS

1. **Khayrul Islam**, Yuwen Zhao, Shen Wang, and Yaling Liu. Machine learning based classification of cells by mechanical properties in microfluidic device. *48th Annual Northeast Bioengineering Conference (NEBEC 2022) (April 2022)*, Columbia University, New York City, New York, 2022
2. **Khayrul Islam**, Tahreen Nabila, and AKM Masud. Investigation of the mechanical properties of polypropylene /carbon nanotube composite by molecular dynamics simulation. *13th International Conference on Mechanical Engineering*, Dhaka, Bangladesh, 2019